Dataset Details

1. Source datasets

	UCI #485 Travel Review Ratings	UCI #484 Travel Reviews (TripAdvisor)
Link	https://archive.ics.uci.edu/dataset/485/tarvel+review+ratings	https://archive.ics.uci.edu/dataset/484/travel+reviews
Origin	Google reviews of attractions across Europe	TripAdvisor reviews, East Asia
Records	5,456 users	980 users
Format	CSV: `User`, 24 category columns (average rating 0-5)	CSV: `User ID`, 10 category columns (average rating 0-4)
Categories	churches, resorts, beaches, parks, theatres, museums, malls, zoo, restaurants, pubs/bars, local services, burger/pizza shops, hotels, juice bars, art galleries, dance clubs, swimming pools, gyms, bakeries, beauty & spas, cafes, view points, monuments, gardens	art galleries, dance clubs, juice bars, restaurants, museums, resorts, parks/picnic spots, beaches, theaters, religious institutions
File in repo	`data/raw/uci485_google_review_ratings.csv`	`data/raw/uci484_tripadvisor_review.csv`

> Note: the brief's "primary link" points to UCI #485 under the name "TripAdvisor Travel Review Ratings", which is the name of UCI #484. We use both. The alternative links (hotel reviews / OULAD) were not used: OULAD is an education dataset, and the hotel review sets need a Kaggle login.

Supporting knowledge base: `data/catalog/destinations.json` is a hand-curated catalog of **32 destinations** (9 regions). Each has 8 preference-dimension strengths, per-person daily cost by budget tier, best months, minimum/ideal/maximum days, and 7-10 costed activities. These are reference estimates, not live prices.

2. Why preprocessing includes simulation

The brief's key fields are `user_id, destination, category, rating, review_text, travel_type, visit_date`. Neither UCI dataset has review text, destinations, travel type or dates; they hold only *user x category average ratings*. They do carry real preference signal from 6,436 travelers, so we build the corpus in three steps.

1. **Category mapping → Traveler DNA.** Source categories are mapped to 8 dimensions (beach, culture, food, nightlife, nature, shopping, wellness, family). For example, culture = mean(churches, theatres, museums, art galleries, monuments). Each traveler's vector is turned into *relative* interest with a z-score then a sigmoid, so it captures what the person likes *more than they usually do*. #484 has no shopping category, so it is imputed from the traveler's overall activity level plus noise.

2. **Trip simulation** (`src/services/dataset_builder.py`, seed 42). For each traveler (plus a second trip for 20%): travel type and budget tier are sampled based on their DNA. The destination is chosen by preference match 75% of the time and at random 25% of the time, which creates the poorly matched trips the model needs as negative examples. Month is in season 65% of the time, and days ≈ the destination's ideal length ± noise.

3. **Rating and review text from one model.** A transparent satisfaction model scores each trip: `3.2 + preference fit + season + budget pressure + duration fit + party fit + N(0, 0.45)`, clipped to 1-

5. The review text is then composed from the *same* factors: activity sentences turn negative when the traveler dislikes that dimension, there are weather complaints off-season, cost complaints on a budget tier at expensive places, and so on. Text, rating and metadata therefore agree, which is what makes semantic retrieval meaningful.

3. Cleaning (applied to any corpus, generated or external)

Dedupe on (user, destination, date, text) · whitespace normalization · drop reviews under 20 characters · coerce rating to an integer 1-5 · normalize `travel_type` to {solo, couple, family, friends, business}.

4. Output corpus: `data/processed/travel_experiences.csv` (7,712 rows)

Field	Example	Notes
review_id	R00099	primary key, used for citations
user_id	G1234 / T56	G = UCI #485 traveler, T = UCI #484 traveler
destination, country, region	Phuket, Thailand, Southeast Asia	
category	beach	dominant dimension of the trip (traveler interest x destination strength)
rating	1-5	distribution: 1★ 43 · 2★ 399 · 3★ 1,872 · 4★ 3,594 · 5★ 1,804 (mean 3.87)
title, review_text	"Great trip", "Spent 5 days exploring..."	60-120 words
travel_type	couple	solo / couple / family / friends / business
visit_date, visit_month	2024-02-18, 2	2019-2025
duration_days	5	2-16
budget_tier, spend_per_day_usd, trip_cost_usd	mid, 94.2, 471	
activities	"Phi Phi Islands speedboat tour | Thai cooking class"	
home_region, source_dataset	South Asia, uci485_google	

Also produced: `traveler_profiles.csv` (6,436 Traveler DNA vectors, used by Travel Twins) and a 200-row sample in `data/sample_data/travel_experiences_sample.csv`.

5. Purpose in the system

Semantic and hybrid retrieval of similar trips · metadata filtering (destination, travel type, rating, budget, month, duration) · preference classification and scoring · Travel Twins collaborative signal · training data for the suitability model · evidence for itinerary citations.

6. Limitations

- Review text is template-composed, so its vocabulary is narrower than real reviews. This is why the cross-encoder adds little on top of hybrid retrieval in our ablation.

- Ratings come from a known function, so the suitability model partly learns that function back (logistic regression scores almost as well as the boosted trees). The pipeline is built to swap in real data: `load_external_experiences()` accepts any CSV with the key fields, and everything downstream is unchanged.